DETAILED DESCRIPTION OF THE INVENTION — POWER MANAGEMENT, ENERGY DELIVERY, AND CHARGING ARCHITECTURES

The intraluminal physiologic monitoring device may incorporate power management and energy delivery architectures configured to support compact operation, safe thermal behavior, and reliable sensing performance across multiple deployment environments.

In various embodiments, energy storage systems may include rechargeable batteries, thin-film batteries, flexible batteries, supercapacitors, hybrid storage assemblies, or externally hosted power modules integrated within pull tab structures or companion hardware accessories.

Power delivery architectures may include regulated voltage rails, dynamic load balancing, low-dropout regulation, buck-boost conversion, duty-cycled sensor activation, and adaptive transmission scheduling designed to optimize battery lifetime while maintaining sensing fidelity.

Charging architectures may include inductive charging, resonant coupling, magnetic contact charging, sealed conductive docking interfaces, capacitive charging systems, or replaceable battery modules. Charging systems may support waterproof integrity, hygienic handling, and alignment through magnetic or mechanical guides.

Certain embodiments may employ distributed power architectures in which internal sensing regions are powered by external energy modules positioned outside the body. Power transfer may occur through flexible conductive traces, embedded wiring, inductive coupling within the pull tab, or other intra-device energy delivery methods.

Thermal monitoring and safety control systems may regulate power usage to prevent excessive heat generation in biologic environments. Power throttling, automatic shutdown, or safe-mode operation may be initiated when thermal thresholds are exceeded.

Energy management algorithms may detect user usage patterns and dynamically adjust sensing schedules, sleep states, or data transmission intervals to optimize operational duration. Devices may support continuous monitoring, scheduled measurement windows, or event-triggered activation.

Future embodiments may incorporate energy harvesting methods such as motion harvesting, thermal gradient harvesting, radiofrequency harvesting, or piezoelectric energy conversion to supplement stored power.

The power management, energy delivery, and charging systems described herein are illustrative rather than limiting. Any architecture capable of supplying energy to intraluminal physiologic monitoring devices while maintaining safe operation is considered within the scope of the present disclosure.